



Lecture Handouts

Semester : 2026-2027 - ODD

Department : AI&DS

Course code with Name : 23CSE24 - Software testing and Automation

Course Faculty : D.Kalaivani,AP/AI&DS

Unit : II - Test Planning

Date of Lecture :

L-2.1

Topics of the Lecture: Test Planning.

Prerequisite Knowledge for complete understanding and learning of Topic: Software Development Life Cycle (SDLC), Software Testing Life Cycle (STLC), Software Quality, Test Process.

Test Planning: Test planning is the process of defining the objectives, scope, strategy, schedule, resources, and responsibilities for software testing.

Goals of Test Planning

1. Define Test Objectives

- Identify the purpose of testing.
- Ensure the software satisfies functional and non-functional requirements.

2. Determine Test Scope

- Specify the features to be tested.
- Identify features that are excluded from testing.

3. Select Test Strategy

- Decide the testing techniques and testing levels.
- Determine whether manual or automated testing will be used.

4. Allocate Resources

- Assign testers and testing responsibilities.
- Arrange testing tools, hardware, software, and test environments.

5. Prepare Test Schedule

- Plan the start and end dates of testing.
- Define milestones for each testing activity.

6. Identify Risks

- Predict possible testing risks.
- Prepare contingency plans to overcome delays and issues.

7. Improve Software Quality

- Detect defects early.
- Reduce software failures after deployment.

Advantages

- Ensures effective use of resources.



- Reduces testing time and cost.
- Improves communication among project members.
- Increases software quality and reliability.

High-Level Expectations:

High-level expectations describe the overall outcomes that the testing process should achieve. They ensure that software meets customer requirements and quality standards.

1. Complete Requirement Coverage

- Every functional and non-functional requirement should be verified through testing.

2. Early Defect Detection

- Defects should be identified during the early stages of software development.
- Early detection reduces maintenance cost.

3. High Software Quality

- Software should be reliable, secure, efficient, and easy to use.
- It should perform correctly under different conditions.

4. Effective Resource Utilization

- Testing should make proper use of available testers, tools, time, and budget.
- Avoid unnecessary testing activities.

5. Risk Reduction

- Identify project risks before software release.
- Minimise the possibility of software failures in the production environment.

6. Clear Communication

- Maintain proper communication between developers, testers, project managers, and customers.
- Report defects and testing progress accurately.

7. Successful Software Release

- Ensure that the software is ready for deployment.
- Deliver a stable and high-quality product to users.

Advantages

- Improves customer satisfaction.
- Reduces software defects.
- Supports better project management.
- Increases confidence in software quality.

Video content / Details of website for further learning (if any):

<https://youtu.be/jic6UgDQcc4?si=7oHVOHEJdSa-TBII>

Important Books/ Journals for further learning including the page nos.

Book Name”Software Testing and Automation” Dr.Monika, D.Rokade – Page No2.3-2.5)



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Topics of the Lecture: Intergroup Responsibilities.

Prerequisite Knowledge for complete understanding and learning of Topic: Software Development Life Cycle (SDLC), Software Testing Life Cycle (STLC), Software Project Management, Test Planning.

Intergroup Responsibilities: It refers to the coordination and communication between different teams involved in software development and testing. Developers, project managers, business analysts, customers, work together throughout the software development life cycle.

Objectives of Intergroup Responsibilities

1. Improve communication among project teams.
2. Ensure proper coordination during testing.
3. Deliver high-quality software.
4. Detect and fix defects quickly.
5. Complete the project within the planned schedule.

Responsibilities of Different Groups

1. Project Manager: The Project Manager is responsible for planning and monitoring the testing process.

Responsibilities

- Prepare the project schedule.
- Allocate resources and budget.
- Monitor project progress.

2. Business Analyst: The Business Analyst understands customer requirements and supports the testing team.

Responsibilities

- Gather customer requirements.
- Prepare requirement specifications.
- Clarify business requirements.

3. Development Team: The Development Team develops the software and fixes reported defects.

Responsibilities



- Develop software modules.
- Perform unit testing.
- Fix reported bugs.

4. Testing Team: The Testing Team verifies that the software meets specified requirements.

Responsibilities

- Prepare the test plan.
- Design test cases.
- Execute test cases.

5. Configuration Management Team: This team controls software versions and testing environments.

Responsibilities

- Maintain software versions.
- Manage source code.
- Prepare test environments.

6. Customer / End User: Customers participate during the final stage of testing.

Responsibilities

- Review software functionality.
- Perform User Acceptance Testing (UAT).
- Verify business requirements.

Advantages of Intergroup Responsibilities

- Improves communication among teams.
- Reduces misunderstandings.
- Detects defects earlier.

Disadvantages

- Poor communication may delay testing.
- Conflicts between teams may affect project quality.
- Lack of coordination increases project cost.

Video content / Details of website for further learning (if any):
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Important Books/ Journals for further learning including the page nos.
Book Name "Software Testing and Automation" Dr.Monika, D.Rokade – Page No(2.6-2.9)



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L-2.3

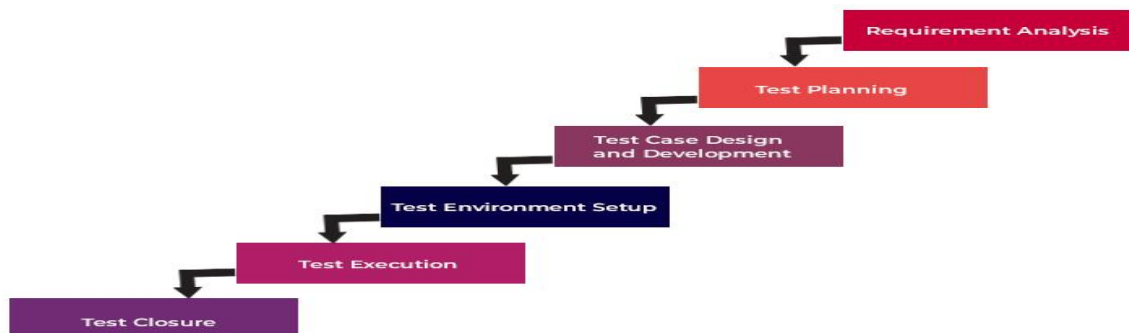
Topics of the Lecture: Test Phases, Test Strategy, Resource Requirements,

Prerequisite Knowledge for complete understanding and learning of Topic: Test Planning, Goal of testing, High level Expectation, Intergroup Responsibilities.

Test Phases: Test Phase are the different stages followed during software testing to ensure that the software is thoroughly tested before delivery. Each phase has a specific objective and helps identify defects at different levels.

Phases of Testing

- 1. Requirement Analysis:** Study software requirements.
- 2. Test Planning:** Define test objectives.
- 3. Test Case Development:** Design test cases.
- 4. Test Environment Setup:** Install hardware and software.
- 5. Test Execution:** Execute test cases.
- 6. Defect Reporting:** Report detected defects.
- 7. Test Closure:** Verify all defects are resolved.



Advantages

- Organised testing process.
- Early defect detection.
- Better software quality.

Test Strategy: It is a high-level document that describes the overall approach, testing methods, objectives, and techniques used for software testing.

It defines **how testing will be performed** throughout the project.

Components of Test Strategy

- 1. Testing Scope:** Features to be tested, Features excluded from testing.
- 2. Testing Levels:** Unit Testing ,Integration Testing , System Testing , Acceptance Testing



3. Testing Types: Functional Testing Performance Testing, Security Testing ,Regression Testing

4. Testing Techniques: Black Box Testing , White Box Testing

5. Entry and Exit Criteria: Conditions to start testing, Conditions to stop testing.

6. Risk Management: Identify project risks, Prepare contingency plans.

Resource Requirements

Resource Requirements refer to the people, hardware, software, tools, and environments required to perform software testing successfully.

Types of Resources

1. Human Resources

- Test Manager
- Test Lead
- Test Engineers
- Automation Testers
- Developers

2. Hardware Resources

- Computers
- Servers
- Mobile Devices
- Network Equipment

3. Software Resources

- Operating Systems
- Browsers
- Databases

4. Testing Tools

- Selenium
- JUnit
- TestNG

5. Test Environment

- Test Server
- Database Server
- Network Configuration

6. Documentation

- Test Plan
- Test Cases
- Bugs Report

Video content / Details of website for further learning (if any):

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Important Books/ Journals for further learning including the page nos.

Book Name”Software Testing and Automation” Dr.Monika, D.Rokade – Page No(2.16-2.23)



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Topics of the Lecture: Tester Assignments.

Prerequisite Knowledge for complete understanding and learning of Topic: Test Planning, Goal of testing, High level Expectation, Intergroup Responsibilities, Test Phases,

Tester Assignments

Tester Assignments refer to the process of allocating testing tasks and responsibilities to members of the testing team.

Tester Assignment Process

1. Analyse Project Requirements
2. Identify Testing Resources
3. Assign Responsibilities
4. Monitor Progress
5. Review Performance

Responsibilities of Test Team Members

1. Test Manager

- Prepare the test plan.
- Allocate testing resources.
- Assign testing tasks.
- Monitor testing progress.
- Review final test reports.

2. Test Lead

- Coordinate testing activities.
- Distribute work among testers.
- Review test cases.
- Verify defect reports.
- Guide the testing team.

3. Test Engineer

- Design test cases.
- Execute test cases.
- Report software defects.
- Perform regression testing.
- Update test results.

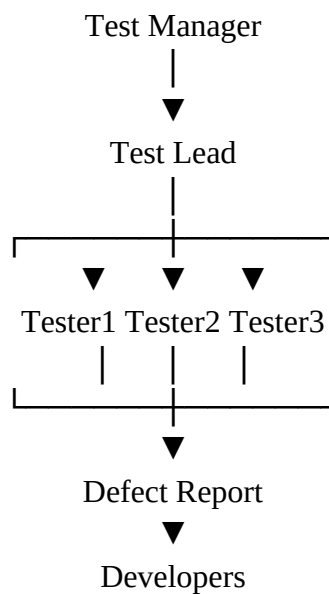
4. Automation Tester

- Develop automation scripts.
- Execute automated tests.
- Maintain automation framework.

5. Developer

- Fix reported defects.
- Perform unit testing.
- Clarify technical issues.

Tester Assignment Diagram



1. Experience

- Assign experienced testers to critical modules.

2. Technical Skills

- Allocate tasks according to programming and testing knowledge.

3. Project Complexity

- Complex modules require skilled testers.

4. Workload

- Distribute work equally among team members.

5. Availability

- Assign tasks based on tester availability.

Video content / Details of website for further learning (if any):
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Book Name”Software Testing and Automation”Dr.Monika,D.Rokade - Page No(2.27-32)

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Topics of the Lecture: Test Schedule, Test Cases.

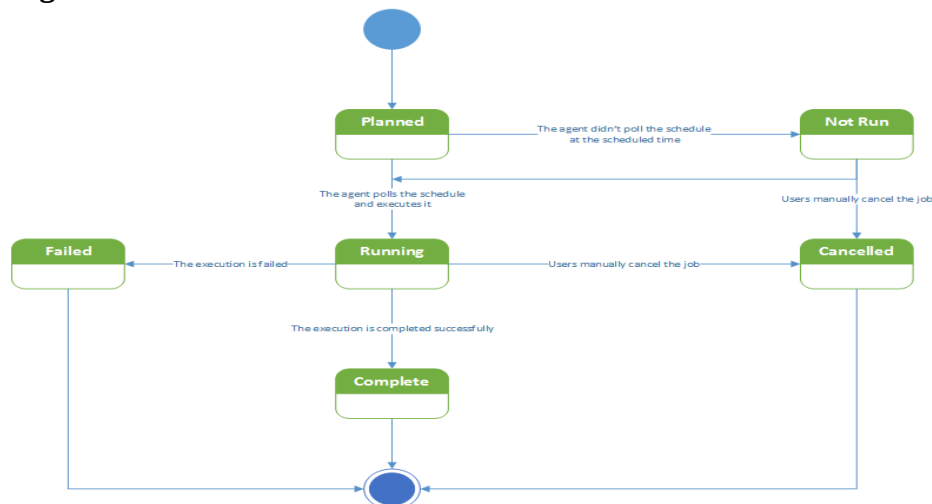
Prerequisite Knowledge for complete understanding and learning of Topic: High level Expectation, Intergroup Responsibilities, Test Phases, Test Strategy.

Test Schedule

Test Schedule is a timetable that specifies **when each testing activity should start and finish**. It helps the testing team complete all testing tasks within the project deadline.

- 1. Test Planning:** Prepare the test plan, Estimate testing effort.
- 2. Test Case Design:** Design and review test cases.
- 3. Test Environment Setup:** Prepare hardware and software, Configure the test environment.
- 4. Test Execution:** Execute all planned test cases, Record test results.
- 5. Defect Reporting:** Report detected bugs, Track defect status.
- 6. Test Closure:** Prepare the final test report, Evaluate testing completion.

Test Schedule Diagram



Advantages

- Improves project planning.
- Ensures timely completion.
- Reduces testing delays.
- Improves software quality.

2. Test Cases

A **Test Case** is a document that contains a set of test inputs, execution steps, expected results, and actual results used to verify whether a software function works correctly.

Components of a Test Case

1. Test Case ID

A unique identification number.

2. Test Objective

Purpose of the test.

3. Preconditions

Conditions that must be satisfied before testing.

4. Test Steps

Actions to perform during testing.

5. Test Data

Input values used for testing.

6. Expected Result

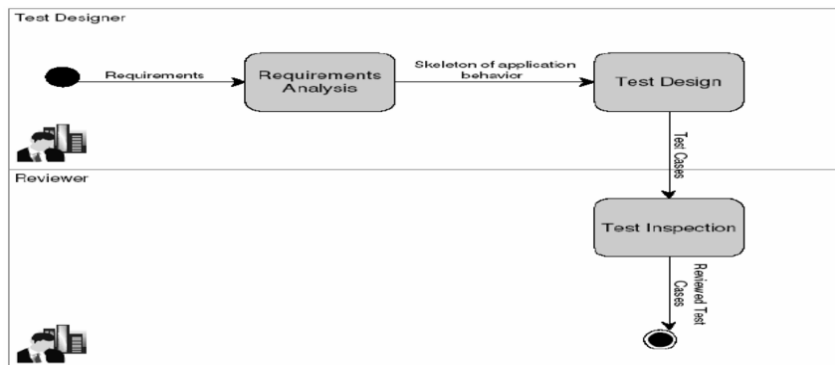
Expected system behaviour.

7. Actual Result

Actual output after execution.

8. Status

Test Case Diagram



Advantages

- Ensures complete test coverage.
- Detects defects early.
- Improves testing accuracy.

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Topics of the Lecture: Bug Reporting, Metrics and Statistics.

Prerequisite Knowledge for complete understanding and learning of Topic: Intergroup Responsibilities, Test Phases, Test Strategy, Test Case, Test Schedule.

Bug Reporting: It is the process of identifying, recording, tracking, and communicating software defects found during testing. A bug report provides complete information about a defect so that developers can understand, reproduce, and fix it.

Bug Reporting Process

1. Identify the Bug

- Execute test cases.
- Detect unexpected behaviour.

2. Record the Bug

- Assign Bug ID.
- Describe the defect clearly.
- Mention severity and priority.

3. Assign to Developer

- Forward the bug report.
- Developer analyses the defect.

4. Fix the Bug

- Developer corrects the defect.
- Updated software build is released.

5. Retest the Bug

- Tester verifies the fix.
- Perform regression testing.

6. Close the Bug

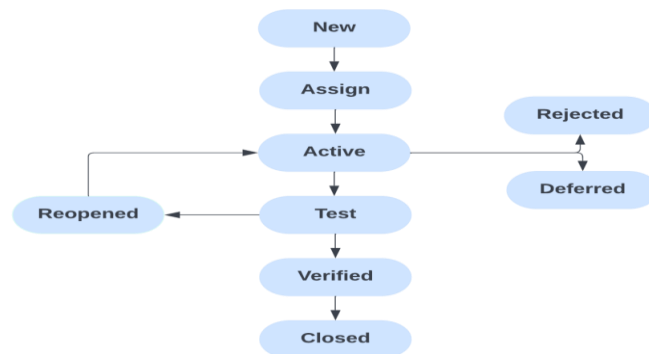
- If fixed successfully, the bug is closed.
- Otherwise, reopen the bug.

Components of a Bug Report

- Bug ID
- Project Name
- Module Name
- Bug Description
- Steps to Reproduce
- Expected Result
- Actual Result
- Severity

- Priority
- Status
- Assigned To
- Reporter Name

Bug Reporting Diagram



2. Metrics and Statistics

Metrics are quantitative measurements used to evaluate the performance and quality of the software testing process.

Statistics are numerical values obtained from testing data that help analyse project progress and software quality.

Common Testing Metrics

1. Test Case Metrics

- Total test cases.
- Executed test cases.
- Passed test cases.
- Failed test cases.

2. Defect Metrics

- Total defects found.
- Open defects.
- Closed defects.
- Reopened defects.

3. Productivity Metrics

- Test cases executed per day.
- Bugs reported per tester.

4. Quality Metrics

- Defect density.
- Test coverage.
- Defect removal efficiency.

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Important Books/ Journals for further learning including the page nos.

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